

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2024

Course: B. Tech

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVC702 & Infrastructure Engineering

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 What is Railway Engineering?	L1	1
a) Railway engineering deals with the design, construction, and operation of railway systems		
b) Railway engineering deals with the design, construction, and operation of highway bridges		
c) Railway engineering deals with the design, construction, and operation of road transportation		
d) None of the mentioned		
2 Which of the following is not a component of the rail?	L1	1
a) Ballast		
b) Foot		
c) Web		
d) Head		
3 Which of the following rail has been standardized for adoption on the Indian railways?	L1	1
a) combination of BH and DH		
b) Flat footed (FF)		
c) double headed (DH)		
d) bull headed (BH)		
4 What is the main purpose of points and crossings in railway tracks?	L1	1
a) Increase the train speed		
b) Facilitate change in direction or track		
c) Improve train stability		
d) Reduce maintenance costs		
5 What is the primary advantage of metro rail systems?	L1	1
a) Reduced construction cost		
b) Reduced road congestion		
c) Faster intercity travel		
d) Increased cargo capacity		
6 Public-Private Partnership (PPP) in metro rail projects is primarily used for:	L1	1
a) Reducing the number of stakeholders		
b) Sharing project risks and investments		
c) Decreasing project completion time		
d) Managing project personnel		

7	Which feature protects harbors from wave action?	L1	1
	a) Jetties	b) Piers	
	c) Breakwaters	d) Wharves	
8	Which factor is critical in the site selection of an airport?	L1	1
	a) Proximity to highways	b) Soil conditions and topography	
	c) Availability of parking	d) Existing railway networks	
9	Which airport component helps manage water runoff?	L1	1
	a) Taxiways	b) Runways	
	c) Airport drainage	d) Terminal buildings	
10	What is the purpose of taxiways in an airport?	L1	1
	a) Facilitate passenger movement	b) Connect runways to terminals	
	c) Store aircraft	d) Serve as emergency runways	
11	Which tunneling method is suitable for hard rock conditions?	L1	1
	a) Shield tunneling	b) Drill and blast method	
	c) Pipe jacking	d) Box pushing	
12	Ventilation in tunnels is required to:	L1	1
	a) Reduce noise levels	b) Remove dust and gases	
	c) Improve structural durability	d) Maintain water drainage	
Q. 2	Solve the following.		12
	A) Explain necessity of Turnout with its components.	L2	6
	B) Define fixtures and fastening. Explain its types.	L1	6
Q.3	Solve the following.		12
	A) Describe types of crossing with neat sketches.	L2	6
	B) Explain Options of Mass Rapid Transit Systems (MRTS).	L2	6
Q. 4	Solve Any Two of the following.		12
	A) Write a short on Inland Water Transport in India.	L1	6
	B) How Harbours are classified based on its utility and location.	L2	6
	C) Define Harbour. Explain components of Harbour.	L2	6
Q.5	Solve Any Two of the following.		12
	A) Define Airport and Airport Engineering. Explain components of airport.	L1	6
	B) Explain factors affecting on site selection of Airport.	L3	6
	C) Explain necessity of Airport Lighting.	L2	6
Q. 6	Solve Any Two of the following.		12
	A) Explain necessity of Tunnel Lining & Tunnel Drainage.	L3	6
	B) Explain necessity of Ventilation and Lighting in tunneling operation.	L3	6
	C) Explain classification of tunnel based on Size and Shape.	L2	6

*** End ***